

Validity Analysis: MENTALCLOUDS

Target context: South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment

Overall risk: **HIGH**

Deployment Context

Use case: Researchers and developers building or evaluating AI-driven counseling support tools assess the capability of LLMs to produce faithful, coherent, and clinically appropriate summaries from counseling dialogues. The benchmark provides a standardized evaluation framework to compare multiple LLMs across summarization quality metrics relevant to the mental health domain.

Target population: NLP researchers and AI practitioners in South Asia and the Middle East and North Africa who are working on applied clinical NLP, digital mental health platforms, and LLM evaluation for low-resource or underrepresented languages and communities.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	2	Significant gaps	high
Output Content 	2	Significant gaps	high
Output Form 	1	Serious concern	high
Average	2.2		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MentalCLOUDS provides a structurally coherent English-only, CBT-aligned counseling-summarization benchmark whose authors are explicit about its American demographic origin [Q70] and three-aspect scope [Q69]. For the SA-MENA practitioner cohort, validity is highly use-case-dependent: South Asian academic researchers benchmarking English LLMs in CBT-aligned Indian institutional settings can derive partial signal, but four of five HIGH-priority dimensions (IO, OO, OC, OF) show

substantive gaps. The output-form dimension is the most severe — ROUGE and BERTScore against English references provide zero valid signal for Arabic, Hindi, Urdu, Bengali, Farsi, or Dari output [Q22, Q23, WEB-8, WEB-10, WEB-12]. The input ontology and output ontology omit constructs (religious framing, shame/honor-protective omission, family-system dynamics, fatalistic attribution) that are clinically salient across the target region [WEB-4, WEB-5, WEB-6, WEB-7] and have not been NLP-operationalized elsewhere. Output content reflects an exclusively India-affiliated annotator pool with confirmed absence of MENA validation [WEB-13]. The benchmark also exhibits a safety-critical gap on suicide-risk capture [Q64] that compounds with documented regional stigma [WEB-1, WEB-2, WEB-3]. The IF dimension (LOWER priority) is the strongest area of fit.

ID	Evidence
Q22	“ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [58] assesses the overlap of n-grams (sequences of n consecutive words) between the generated summary and reference summaries.” (p.9)
Q23	“This metric measures the number of overlapping units such as n-gram, word sequences, and word pairs between the generated summary evaluated against the gold summary typically created by humans.” (p.9)
Q64	“Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified.” (p.15)
Q69	“Finally, this work explored only three aspects (counseling component) of the conversation.” (p.16)
Q70	“Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics.” (p.16)
WEB-1	BMC Psychiatry 2020 systematic review (n=6,767) — stigma toward mental illness in India higher than Western countries (https://link.springer.com/article/10.1186/s12888-020-02937-x)
WEB-2	Pakistan NPMS 2022 — ~1 in 3 adults meet psychiatric criteria; help-seeking low partly due to stigma (https://www.medrxiv.org/content/10.1101/2024.07.08.24310056v1.full.pdf)
WEB-3	Bangladesh 2023 — mental health treatment gap ~94%; stigma a primary structural barrier (https://pmc.ncbi.nlm.nih.gov/articles/PMC10579681/)
WEB-4	TIIP framework — Islamic psychology clinical taxonomy not represented in the annotator pool's training (https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864)
WEB-5	Keshavarzi & Haque four soul aspects framework — additional MENA clinical normative reference (https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000)
WEB-6	Ruqyah, dhikr, salah documented as cultural/spiritual treatment adjuncts in Muslim mental health literature (https://pmc.ncbi.nlm.nih.gov/articles/PMC8267770/)
WEB-7	Narrative review notes CBT's directive stance is documented as compatible with Arab and South Asian Muslim clinical expectations in some contexts (https://pmc.ncbi.nlm.nih.gov/articles/PMC9119255/)
WEB-8	arXiv 2602.02440 — Arabic LLM evaluation: BLEU ~0.54 vs LaBSE ~0.78 confirms n-gram metrics like ROUGE poorly suited for Arabic clinical NLP (https://arxiv.org/html/2602.02440)
WEB-10	ACM Web Conference 2024 (XLEval) — statistically significant BERTScore gap between English (0.9206) and Hindi for healthcare queries, confirming metric validity concerns for non-English clinical NLP (https://dl.acm.org/doi/10.1145/3589334.3645643)
WEB-12	AraHealthQA 2025 — BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA (https://arxiv.org/html/2508.20047v2)
WEB-13	JMIR Mental Health 2024 — confirms no MENA-based mental health professional validation of MentalCLOUDS annotations conducted or announced; expert pool exclusively India-affiliated (https://mental.jmir.org/2024/1/e57306)

Practical Guidance

What This Benchmark Measures

MentalCLOUDS measures, for English-language LLMs, the ability to produce CBT-aligned aspect-based summaries (Symptom and History, Patient Discovery, Reflecting) of American-sourced counseling dialogues, as judged by ROUGE/BERTScore overlap with India-annotated English references and by an India-based expert panel applying the Sekhon six-parameter clinical-acceptability framework plus a hallucination scale. Its strongest contribution for the SA-MENA cohort is the hallucination evaluation framework and the comparative open-source <7B model leaderboard, both of which transfer across

regions as language-agnostic constructs (IF dimension is the strongest area of fit).

Construct Depth

Construct depth is shallow-to-moderate for the SA-MENA target context. The benchmark probes only three counseling aspects [Q69] and does not surface psychotherapy-type identification [Q63], suicide-risk capture [Q64], religious/family-system content, or non-English language behavior. The IO and OO dimensions miss key region-relevant categories; the OF dimension provides no valid cross-lingual signal; the OC dimension lacks MENA professional validation [WEB-13]. South Asian English-system evaluation is the only sub-population for which the benchmark provides moderately complete construct coverage.

What Else You Need

For MENA Arabic/Farsi/Dari practitioners (HIGH priority for OF, OO, OC): an entirely separate evaluation infrastructure is needed — multilingual reference summaries, cross-lingual metrics (LaBSE, XLM-R BERTScore, AraBERT-based BERTScore [WEB-8, WEB-9, WEB-11]), and a MENA expert annotator pool with Islamic psychology training [WEB-4, WEB-5, WEB-16]. For South Asian Hindi/Urdu/Bengali system builders (HIGH priority for IO, OO, OF): regionally adapted counseling component taxonomies covering izzat, family dynamics, and stigma-protective omissions, plus Hindi/Urdu reference summaries and MuRIL/IndicBERT-based metrics. For all sub-populations: additional safety-focused evaluation of suicide-risk capture given the documented benchmark failure [Q64] compounded by regional stigma [WEB-1, WEB-2, WEB-3].

ID	Evidence
Q63	"The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)
Q64	"Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)
Q69	"Finally, this work explored only three aspects (counseling component) of the conversation." (p.16)
WEB-1	BMC Psychiatry 2020 systematic review (n=6,767) — stigma toward mental illness in India higher than Western countries (https://link.springer.com/article/10.1186/s12888-020-02937-x)
WEB-2	Pakistan NPMS 2022 — ~1 in 3 adults meet psychiatric criteria; help-seeking low partly due to stigma (https://www.medrxiv.org/content/10.1101/2024.07.08.24310056v1.full.pdf)
WEB-3	Bangladesh 2023 — mental health treatment gap ~94%; stigma a primary structural barrier (https://pmc.ncbi.nlm.nih.gov/articles/PMC10579681/)
WEB-4	TIIP framework — Islamic psychology clinical taxonomy not represented in the annotator pool's training (https://www.routledge.com/Applying-Islamic-Principles-to-Clinical-Mental-Health-Care-Introducing-Traditional-Islamically-Integrated-Psychotherapy/Keshavarzi-Khan-Ali-Awaad/p/book/9780367488864)
WEB-5	Keshavarzi & Haque four soul aspects framework — additional MENA clinical normative reference (https://www.tandfonline.com/doi/abs/10.1080/10508619.2012.712000)
WEB-8	arXiv 2602.02440 — Arabic LLM evaluation: BLEU ~0.54 vs LaBSE ~0.78 confirms n-gram metrics like ROUGE poorly suited for Arabic clinical NLP (https://arxiv.org/html/2602.02440)
WEB-9	MDPI Healthcare 2025 — AraBERT, CAMELBERT, MARBERT applied to Arabic mental health classification; BERTScore for Arabic clinical summarization not yet evaluated (https://www.mdpi.com/2227-9032/13/9/985)
WEB-11	MDPI Applied Sciences 2025 — XLM-RoBERTa BERTScore demonstrated robust for cross-lingual summarization including low-resource languages; not yet applied to clinical mental health summarization (https://www.mdpi.com/2076-3417/15/14/7800)
WEB-13	JMIR Mental Health 2024 — confirms no MENA-based mental health professional validation of MentalCLOUDS annotations conducted or announced; expert pool exclusively India-affiliated (https://mental.jmir.org/2024/1/e57306)
WEB-16	Stanford Muslim Mental Health & Islamic Psychology Lab — institutional locus for Islamic psychology NLP (https://med.stanford.edu/mmhip.html)